Customer Platform Choice Prediction

A Shared-Encoder Neural Network

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Abstract

This study predicts the online platform selected by a customer from three alternatives, given customer attributes and platform-specific delivery times and costs. Using 7,000 choice scenarios, we compare a multinomial logistic-regression benchmark with a compact neural network that applies the same scoring function to each platform. The analysis covers exploratory statistics, alternative-level feature construction, cross-entropy optimization, and a small search over Adam hyperparameters. The shared-encoder model reaches approximately 70% validation accuracy in the original experiments and illustrates how nonlinear choice models can capture trade-offs between cost and delivery time.

1 Data and exploratory analysis

Each observation records two customer attributes (`GROUP` and `MEMBER`), the delivery time and cost of platforms A, B, and C, and the selected alternative (`CHOICE` $\in \{0, 1, 2\}$). The dataset consists of 7,000 observations and eight explanatory fields, excluding the target.

Descriptive statistics and distributional plots indicate substantial differences between alternatives. Platform A has a particularly wide cost distribution, with a mean near 89 and observations above 1,200, while the cost distributions for B and C are more concentrated. Delivery times also vary appreciably: mean delivery time is about 87 minutes for A versus 18 minutes for B. Boxplots highlight the importance of accounting for scale differences and outliers. Pairwise plots suggest that cheaper and faster platforms are more often selected.

2 Feature representation and models

2.1 Input construction

For each customer, we repeat the two customer-level attributes across the three alternatives and append each platform's delivery time and cost. This creates an input tensor

$$X \in \mathbb{R}^{7000\times 3\times 4}, \qquad y \in \{0, 1, 2\}^{7000}. \tag{1}$$

Each row of $X_i \in \mathbb{R}^{3\times 4}$ describes one platform. The original modeling workflow used PyTorch tensors and mini-batches of 100 observations.

2.2 Multinomial logistic regression

A logistic regression on the flattened 12-dimensional representation, using z-score standardization and the L-BFGS solver, provides a simple interpretable reference model. The exploratory benchmark achieved 66.9% classification accuracy when evaluated on the full dataset used for fitting; this is an in-sample metric rather than a held-out estimate.

2.3 Shared-encoder neural network

The neural network applies a single multilayer perceptron $g_\theta : \mathbb{R}^4 \to \mathbb{R}$ to each alternative, with hidden widths of 32 and 16 and ReLU activations:

$$\boldsymbol{z}_i = \big(g_\theta(X_{i,A}),\ g_\theta(X_{i,B}),\ g_\theta(X_{i,C})\big), \qquad p_{ik} = \frac{e^{z_{ik}}}{\sum_{j\in\{A,B,C\}} e^{z_{ij}}}. \tag{2}$$

The three scores become class logits and are trained using cross-entropy loss. Weight sharing gives the model *permutation equivariance*: reordering the alternatives reorders their scores and probabilities in the same way, without changing the underlying scoring rule.

3 Training and model selection

We train with the Adam optimizer and a mini-batch size of 100. Five configurations are compared, varying learning rate, Adam momentum parameters, and weight decay. The learning rates considered are 0.0018, 0.0020, and 0.0022; the two pairs of momentum parameters are $(0.9, 0.98)$ and $(0.92, 0.985)$; weight decay values include 0, 2×10^{-5}, and 5×10^{-5}.

Training proceeds for up to 100 epochs with early stopping. In the recorded hyperparameter comparison, the configuration with learning rate 0.002, betas $(0.9, 0.98)$, and no weight decay achieved the highest observed validation accuracy, approximately 69.2%. Earlier exploratory runs reached approximately 72%, suggesting some variation across training procedures and model selection runs. The loss fell to roughly 0.73 as training progressed.

4 Results and interpretation

The trained shared-encoder network captures nonlinear associations between platform attractiveness and customer choice. The original analysis found validation accuracies around 69–72%; the model performed particularly well when given sufficient time to learn from delivery and cost differences. Its shared weights also reduce the number of free parameters and impose a consistent treatment of the three platforms.

A further check exchanges platforms A and B in the same input observations. Their predicted probabilities exchange positions, while the third platform's probability is preserved. This behavior confirms the intended structural property of the model.

The reported 66.9% logistic benchmark is measured in-sample, while the neural-network results are validation accuracies. Those original figures should therefore not be interpreted as a controlled out-of-sample comparison. Reliable model comparison requires applying both estimators to the same held-out data, as implemented in the accompanying notebook.

5 Conclusion

This project demonstrates a compact neural-network formulation of customer platform choice, integrating customer characteristics with platform delivery and pricing attributes. A shared MLP encoder provides a natural representation of alternative-level trade-offs and supports straightforward verification of permutation equivariance. Further improvements could include richer behavioral covariates, more extensive model selection, and benchmarking on additional held-out datasets.